

Operation and Maintenance Manual

Translation of the Original Operating Manual

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Gas-Gun Setup for Impact Loading



Newtomech

“Best solutions for your success”

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Gas Gun for Impact Testing as per ASTM D8101/8101M

Introduction:

An impact test is performed by accelerating a defined projectile to a specified velocity, typically with the use of a single-stage gas gun, into a composite test panel that is supported in a fixture. The test panel is supported in a circular fixture with precision bolts extending through a front clamp and the specimen itself to avoid slipping of the specimen at the boundaries. The location of the holes is remote from the impact site so that damage is not initiated at the holes.

Depending on the kinetic energy of the projectile, it may or may not damage or penetrate the test panel. The penetration resistance is quantified by either the velocity or kinetic energy required to penetrate the test panel. The penetration resistance is a function of the geometry and materials of the test panel. Comparisons between materials or material conditions cannot be made unless identical test configurations and test conditions are used. Procedures and equipment for measuring the impact velocity and orientation of the projectile just before impact are required. Equipment for measuring the residual velocity of the projectile after penetration, if it occurs, is desirable but not required.


Scope:

1. Impact testing of target plate as per ASTM D8101/8101M
2. Predicting Kinetic energy of the projectile before impact on target plate
3. Calculating energy absorption of target plate from difference in kinetic energy of projectile before and after penetration

Technical specifications:

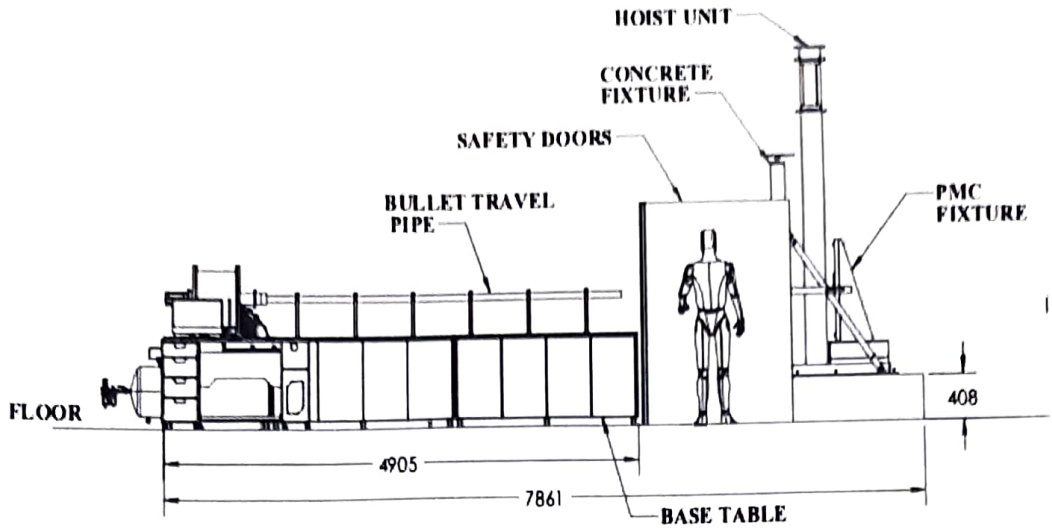
Sl. No.	Specification	
1.	Max. Chamber Design Pressure	30 bar
2.	Gas gun chamber capacity Maximum	60 litres
3.	Striking mass	10-50g for polymers and another mass up to 2.6 kg to test concrete specimen
4	Stand for mounting gas chamber and barrel	1m from ground
5.	Projectile velocity measurement	0.01m/s accuracy Laser Diode System
6.	Pneumatic system Control	Solenoid Valve for pressurised air charging in Chamber and triggering system
7.	Pressure gauge to measure chamber pressure	Analogue type of range 0-50 bar gauge pressure



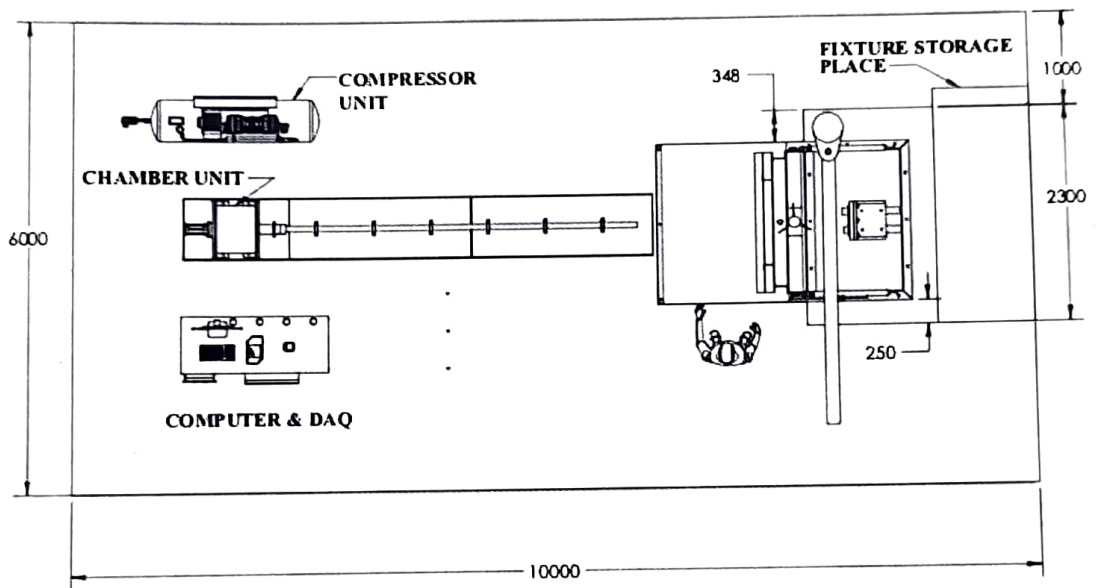
8.	Fixture for fixing target	As per ASTM D8101M for polymer and another fixture to hold 1.5 m x 1.5 m x 80 mm thick concrete specimen with clamped boundary condition on 4 edges.
9	Projectile	As per ASTM D8101M for polymers and 50 mm dia, 300 mm long projectile for concrete specimen.
10	Accelerometer to measure Impact response	100,000 g shock accelerometer of PCB or reputed make of Piezo type
11	Data Acquisition System	MODEL NO: IV 48B5B39 (2 CHANNEL USB SIGNAL CONDITIONER)
12.	Safety	Pressure overload trip facility
13.	Computer System	Windows OS, i7 processor, 16 GB RAM and 500 GB Hard drive. Monitor 15", keyboard, mouse and UPS shall be supplied.



Schematic Diagram



Front View



Top View



Pneumatic Control System:

Pneumatic control is used for charging of air in the chamber and releasing of air to the barrel to accelerate the projectile. Figure 5 shows the pneumatic circuit used in the control panel.

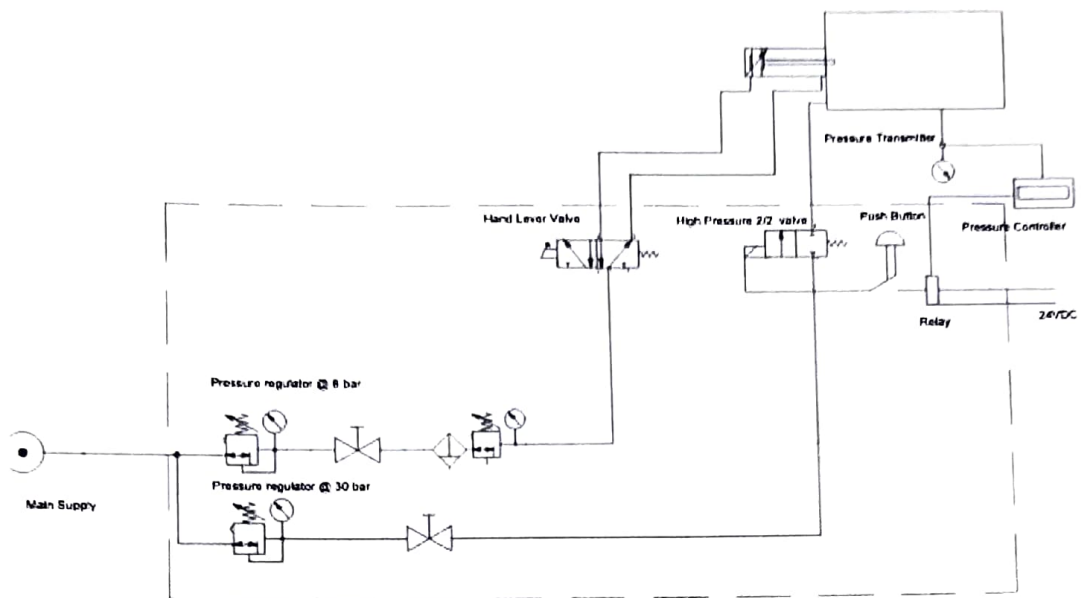


Figure 4 Pneumatic circuit

**Operation:**

- Step 1 :** Compressor to be switched on to charge air pressure in the received
- Step 2:** Target plate to be fixed in the fixture. Screws should be tightened with uniform torque.
- Step 3:** Target guard to be closed without gap between target plate and guard and lock using toggle clamp.
- Step 4:** Pneumatic control panel lever should be kept in charging position
- Step 5:** Projectile to be placed inside the barrel as shown in Figure. Place metal cap on the slot and tighten the using T bolt.
- Step 6:** Ensure the target guard closure and closure of projectile receiving box at rear side of the target mounting angle plate.
- Step 7:** Charge the air in the chamber using pressure controller till reaching test air pressure in the chamber. Monitor air pressure through pressure gauge.
- Step 8:** Check the data acquisition system and trigger impact response sensor(accelerometer)
- Step 9:** Trigger the triggering lever for accelerating the projectile to impact on target.
- Step 10:** Save the acceleration Vs Time recorded data in the computer and note the projectile velocity. Dismantle the target plate.

**Maintenance:**

1. **Compressor should be maintained as per Compressor manufacturer's instructions.**
2. **Gas gun should be maintained as per below:**

Daily Maintenance:

- Gas gun should be cleaned for dust and dirt
- Oil level in FLR unit of Pneumatic Control unit to be monitored. If there is decrease in oil level less than minimum level, recapping should be done.
- Check for leakage of air if any.



Safety Precautions:

- The following notation used in this manual is to provide precautions required to ensure safe usage of Gas gun operator and Gas gun controller.
- The safety precaution that are provided are extremely important to ensure safety. Always read and heed the information provided in all safety precautions.

 WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury. Additionally, there may be severe property damage.
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 Caution	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury, or property damage.
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Control panel information

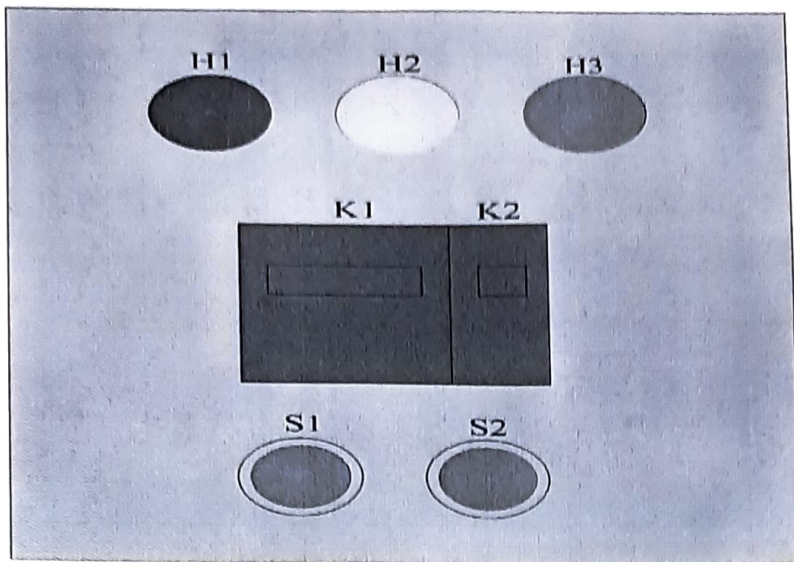
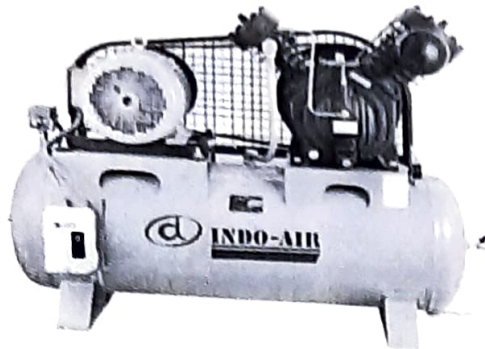


Figure 4 Control Panel

- H1, H2, & H3 will indicate the R, Y, & B terminals.
- The K1 and K2 are trippers which is used to switch on the control panel supply, in which K1 is a three-phase tripper.
- K1 is used to switch ON the power supply of compressor motor.
- K2 is used to switch ON the power supply of pneumatic control.
- S1 and S2 are indicators of K1 and K2 respectively.



Air compressor



MODEL	MOTOR		NO. OF CYL.	RPM	RECEIVER		MAX PRESSURE		PISTON DISPL		
	HP	KW			CFT.	LTRS.	PSIG	KG/CM ²	CFM	M ³ /M	LTR/MIN
IA 10 NL	3	2.2	2	660	7.95	225	142	10	13.20	0.374	374

- Sealed and high temperature resistant pre-lubricated bearings
- Well designed cooling system of air cooled inter/after coolers
- Special material piston ring which retains its properties under high temperature working conditions
- Piston ring outer surface always in contact with the cylinder bore even after thousands of hours of running and corresponding wear ensuring unchanged pumping up time
- Special material piston ensures the retention of cylinder bore finish in adverse temperature condition and expansion of piston